Alessio Olivieri

Artificial Intelligence | Robotics | Machine Learning | Systems Engineering

Rome, Italy | olivieri.1973323@studenti.uniroma1.it | LinkedIn | GitHub | Portfolio

Profile

MSc student in **Artificial Intelligence & Robotics** at Sapienza University of Rome, combining research in **deep learning, interpretable vision models, GPU computing, and cybersecurity ML** with practical open-source and software engineering. Interested in research-oriented AI and systems roles.

Education

MSc, Artificial Intelligence & Robotics – Sapienza University of Rome In progress
Focus: machine learning, robotics, computer vision, optimization, neural networks, and HPC.

BSc, Applied Computer Science & Artificial Intelligence – Sapienza Completed | 101/110
Focus: algorithms, statistics, linear algebra, software engineering, networking, and AI.

Experience

Mathematics Tutor – Linear Algebra, Sapienza University of Rome Sep 2025 – Feb 2026
Delivered supplementary Linear Algebra classes alongside Prof. Max Tschaikowski's lectures, supporting bachelor's students in Applied Computer Science and Artificial Intelligence (ACSAI).

Programming Tutor Nov 2023 – Jan 2024
Tutored university students in programming and technical subjects; developed learning materials and communicated feedback.

Plum Juice – Technical Writer & Team Member Sep 2024 – present
Wrote technical articles and contributed to the team's social media presence.

Research & Technical Projects

Vision Transformer Pruning and Interpretability. Implemented ACDC-style causal analysis and activation patching in PyTorch; removed **85% of attention heads**, enabling about **2× faster CPU inference** after retraining with a 9% accuracy drop.

Swing AllReduce – GPU Collectives (Bachelor's Thesis). Implemented and benchmarked NCCL/MSCCLang multi-GPU communication, including Leonardo HPC testing; observed topology-dependent speedups in tested configurations.

Encrypted Network Traffic Classification – Research. Studied graph-based traffic classification with controlled ablations and capture-provenance audits; identified limits of flow-disjoint generalization claims.

Gadgetbridge – Samsung Gear S3 Interoperability. Developed Android–Tizen Bluetooth/SAP integration, notification replies, reconnection fixes, log analysis and regression tests on physical devices.

Video Action Segmentation. Compared MS-TCN and bidirectional GRU architectures for frame-level segmentation; best reported macro-F1: **0.83**.

Real-Time Fire Detection. Optimized a surveillance-camera detection pipeline with NumPy and background subtraction from **0.1 to 60 FPS** in project benchmarks.

LLM-to-SQL Assistant. Built a Streamlit application translating natural-language questions into SQL queries using Llama models.

Parallel Neural Network. Built an MLP in C/Pthreads and CUDA, reporting about **10× speedup** on 10 CPU cores.

Other software/data projects. WasaPhoto RESTful social platform (Go, Vue, SQLite, Docker) and ingredient-network analysis of over 6,000 scraped recipes.

Technical Skills

Programming: Python, C, Java, Go, JavaScript, SQL, R; PyTorch, NumPy, CUDA, Pthreads.
Tools & domains: Linux, Docker, Git, Android, Bluetooth, NCCL/MSCCLang, ML evaluation, computer vision, distributed systems.
Languages: English (C2), Italian (C2), Spanish (A2), Polish (A2).